2022A F=ma Exam: Problem 20

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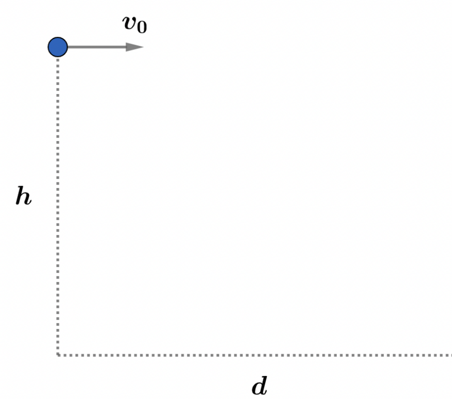

The shape of the wire for which there is no normal force on the bead is the same as the trajectory of the bead in the absence of the wire i.e. projectile motion. From kinematics, the time it takes for the bead to reach the ground is

$$h = \frac{1}{2}gt^2$$
$$t = \sqrt{\frac{2h}{g}}$$

The horizontal displacement is

$$d = v_o t = v_0\sqrt{\frac{2h}{g}}$$

so the answer is $\boxed{\text{B}}$.